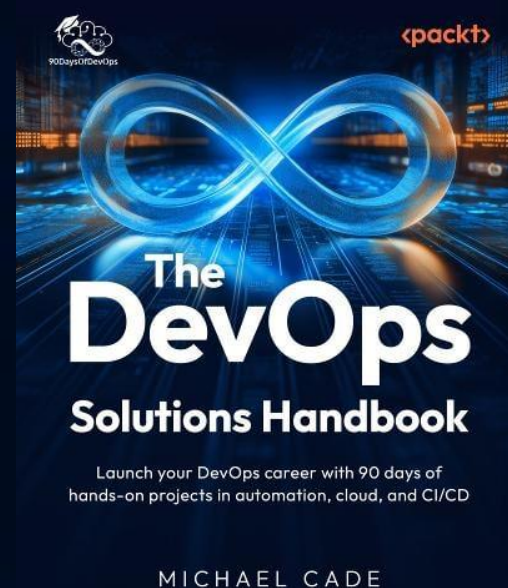


The Hypervisor Hunger Games:

Protecting VMs, DevOps Tooling,
and State via GitOps

Michael Cade
Field CTO | Technologist
#90DaysOfDevOps

September 24, 2026



Kubernetes as the New Infrastructure OS

One platform for containers and
VMs running side-by-side

Broadcom has forced the industry to
reconsider where VMs run

Treating data protection as an
afterthought creates operational risk





Kubevirt 101



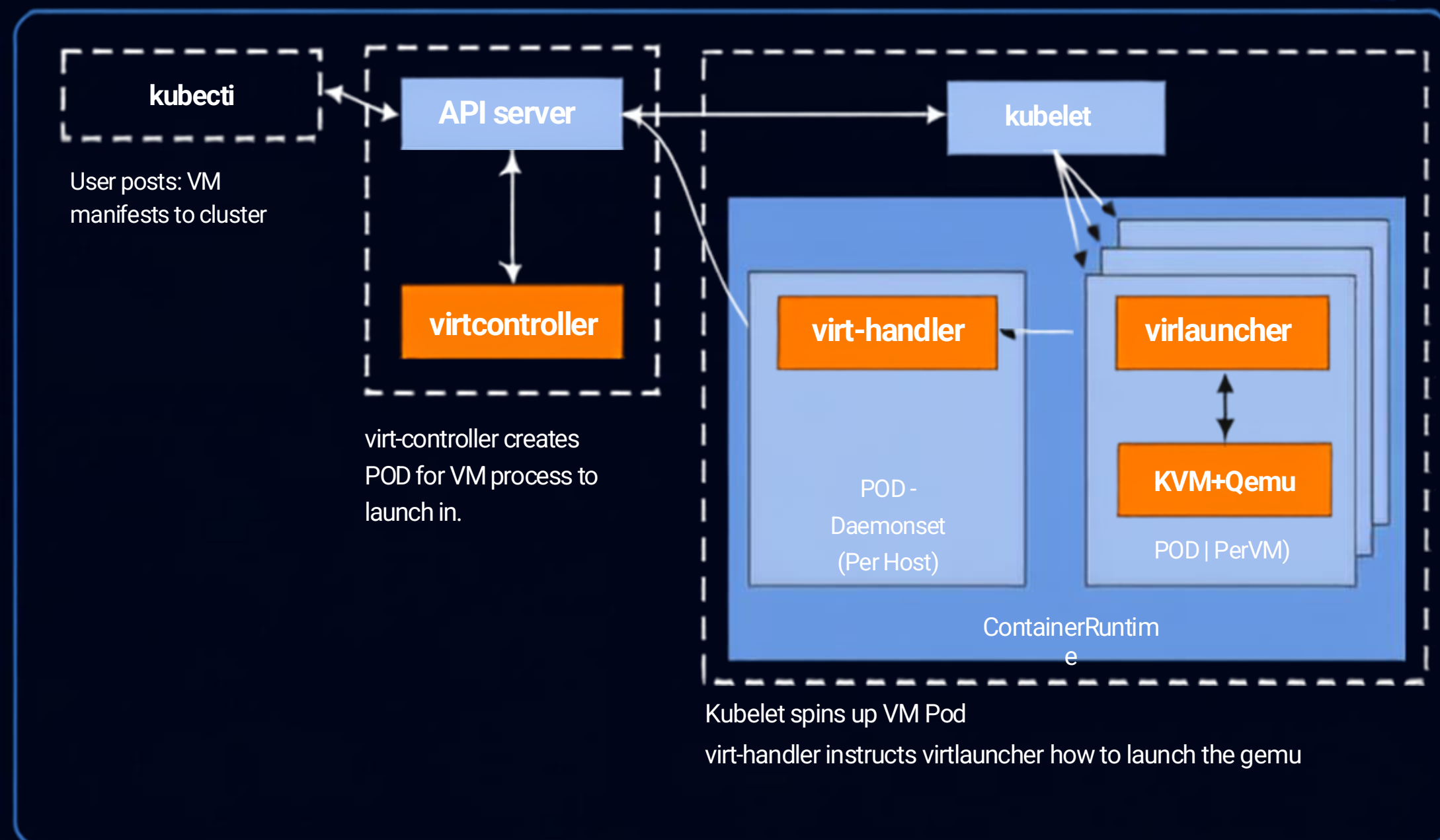
Kubernetes becomes the hypervisor



A Kubevirt VM is a KVM+qemu process running inside a pod.



KubeVirt is a Kubernetes extension that allows running traditional VM workloads natively side by side with Container workloads.





VM Storage



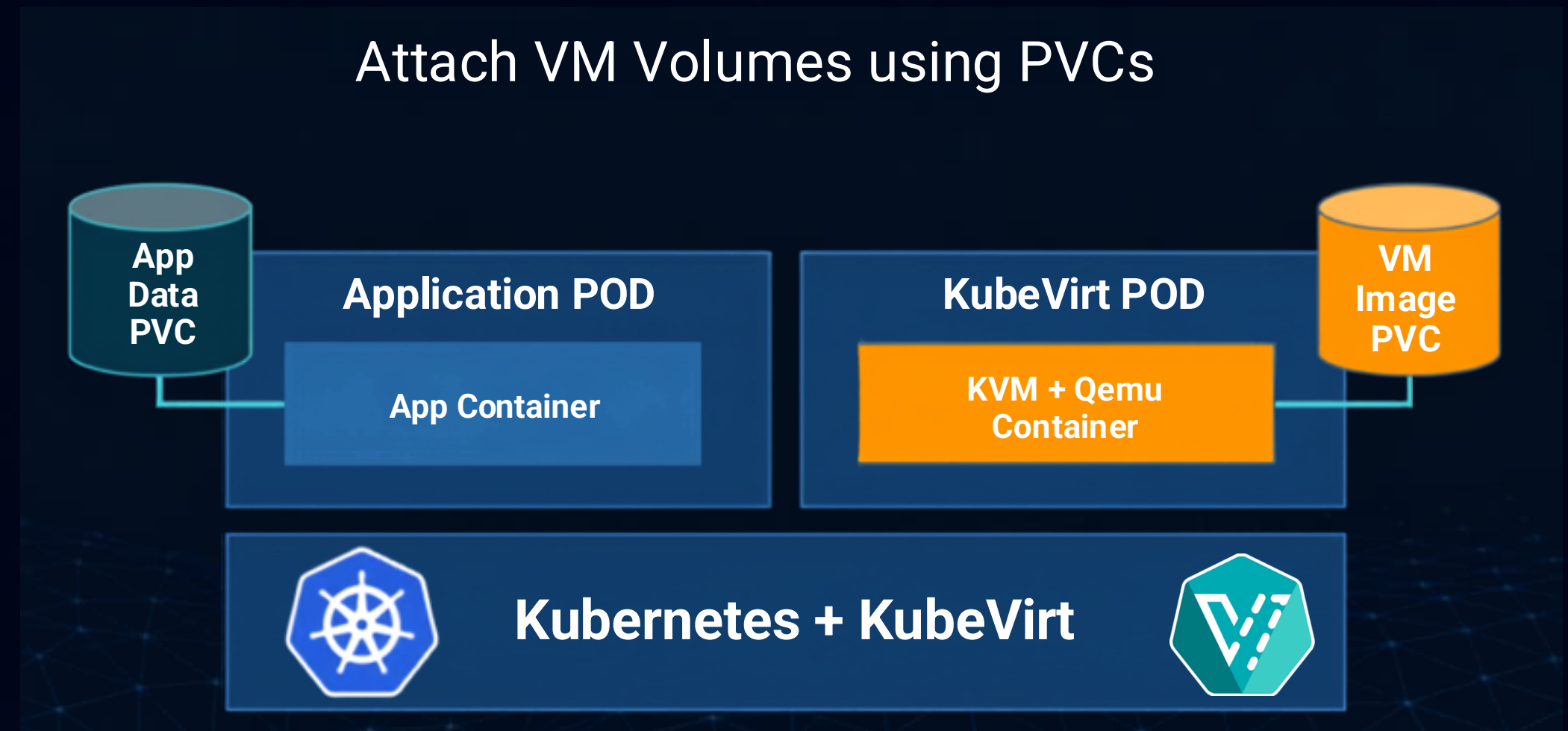
The rise of stateful workloads in Kubernetes



The storage in KubeVirt is used in the same way as the pods



Every VM, has an OS and probably additional data disks





Data Protection: The Common Afterthought



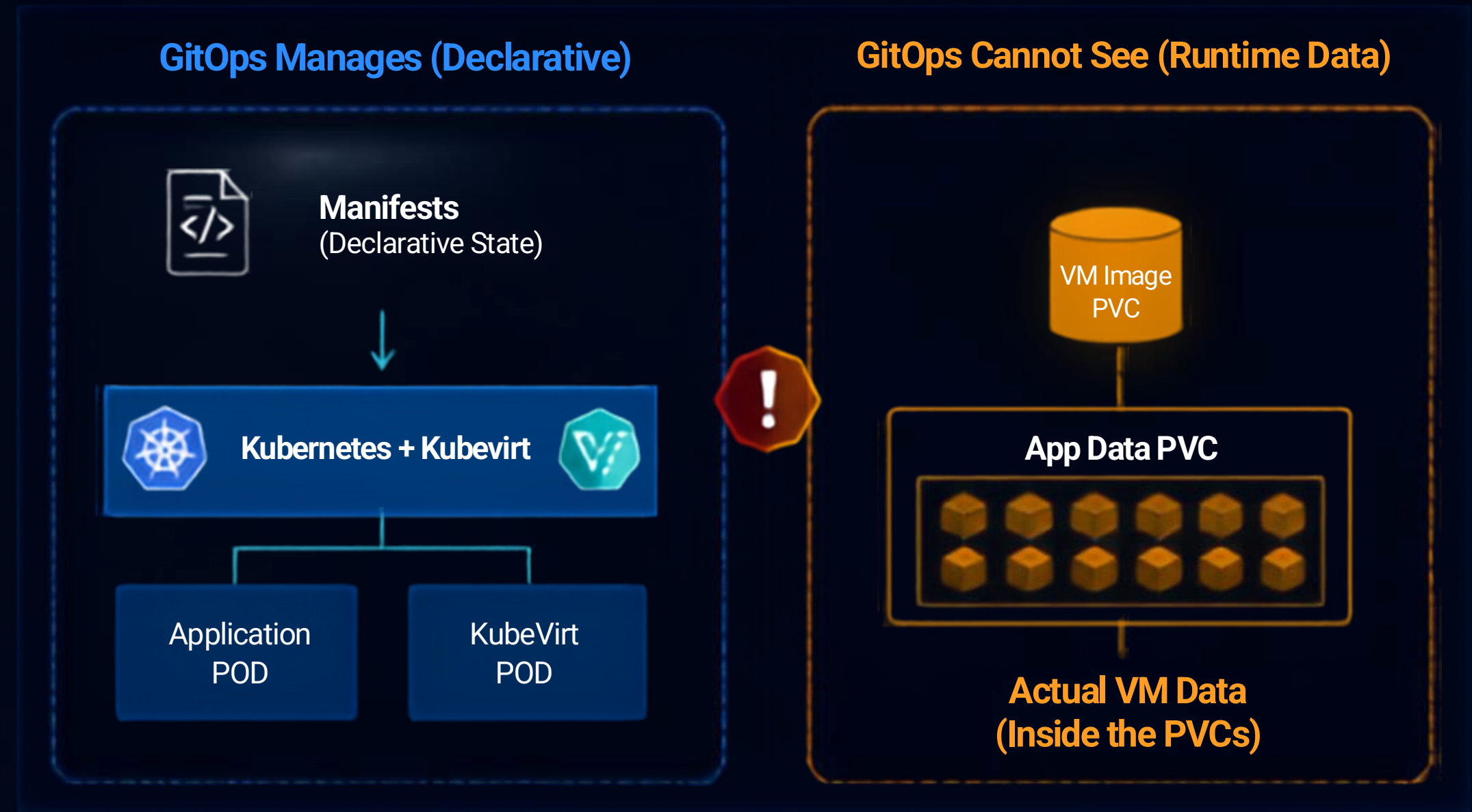
A VM is a stateful workload.



GitOps can manage and rebuild what it knows about, but it cannot see the contents inside the VM.



Much like a containerised database, the real gap is inside the PVCs.



Protect the contents of the PVCs, **not just the manifests.**

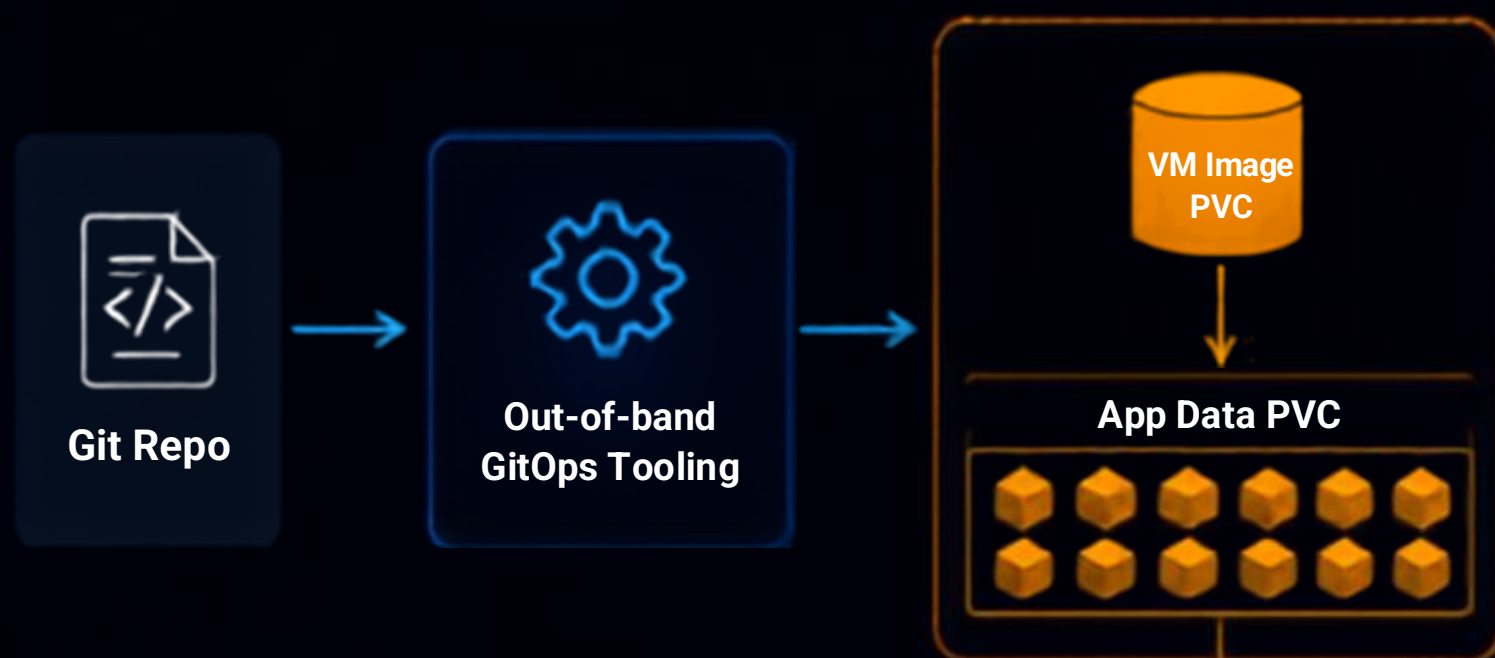


We have 2 choices

1

Protect outside GitOps

Use an out-of-band GitOps process to protect these VMs.

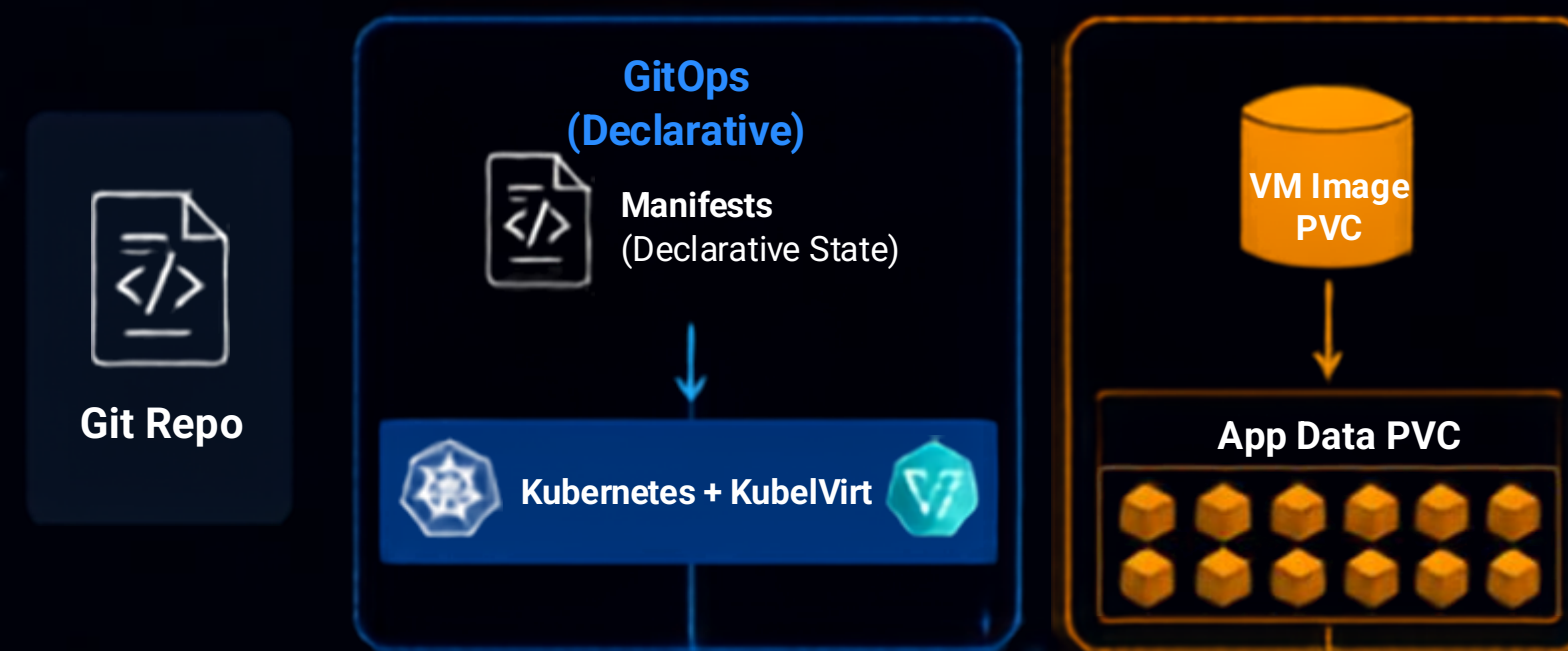


VS

2

Protect within GitOps

Combine Kubernetes-native VM protection with our GitOps workflows.



Application Consistency matters

If the VM is running a database, how do we ensure the database backup is clean and consistent?



A GitOps-Driven Data Protection Strategy

Treat backup infrastructure entirely as code within GitOps pipelines

Use declarative, immutable data protection policies integrated via ArgoCD

Handle large VM data footprints alongside container state seamlessly

Safeguard DevOps tooling registries with automated, version-controlled protection

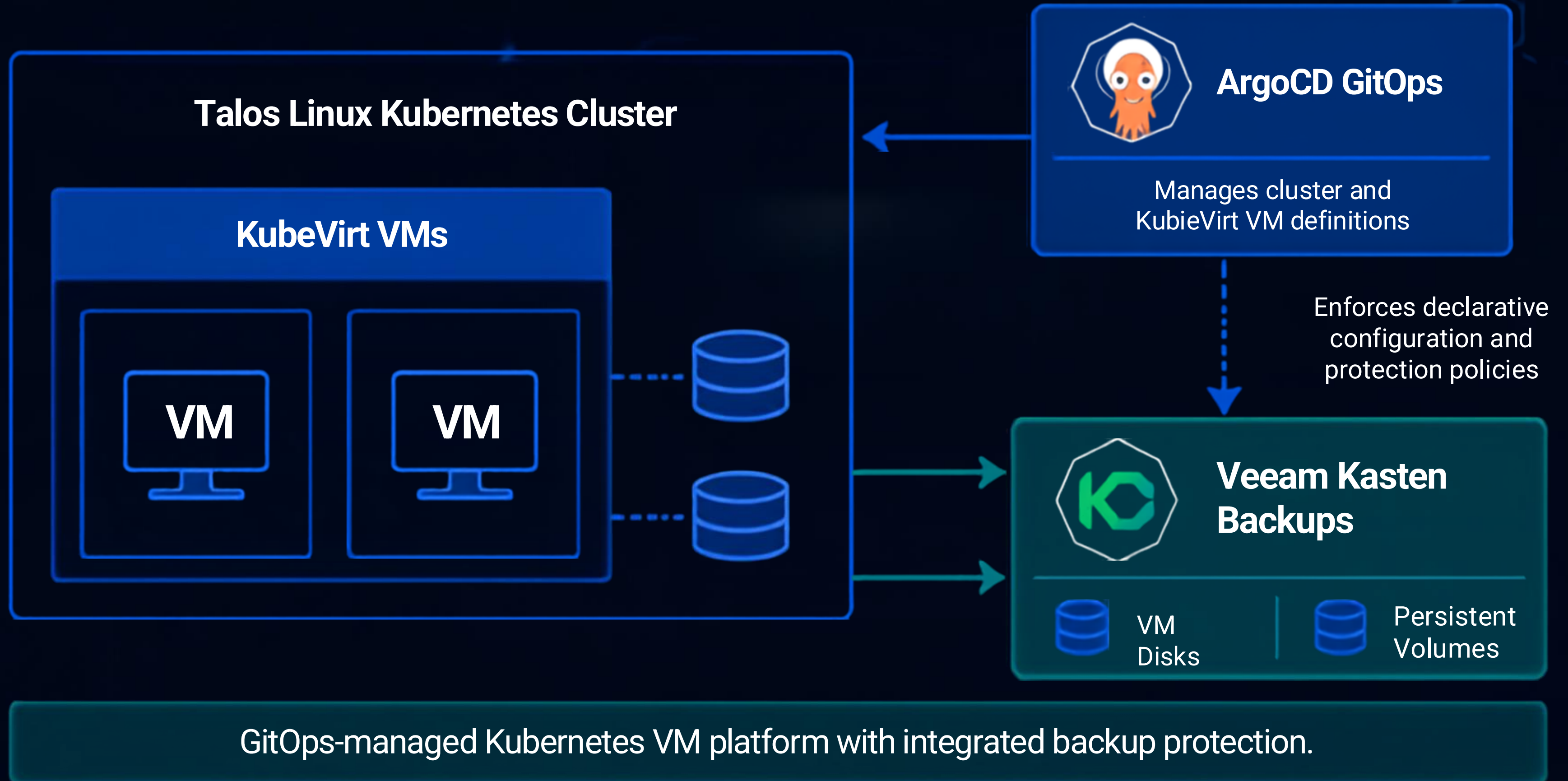
Automate multi-cluster recovery workflows for fast, reliable restoration

Integrating backups into your GitOps Pipeline



Declarative protection policies automate resilience across virtual and container workloads

Demo



☐ Syncing 0

APPLICATIONS TILES

[Log out](#)

Sort: name ▼ Items per page: 10 ▼

kasten-config

Project: default

Labels:

Status: ♥ Healthy ✓ Synced

Repository: <https://github.com/MichaelCade/kubevirt-kasten-...>

Target Revision: main

Path: kasten/config

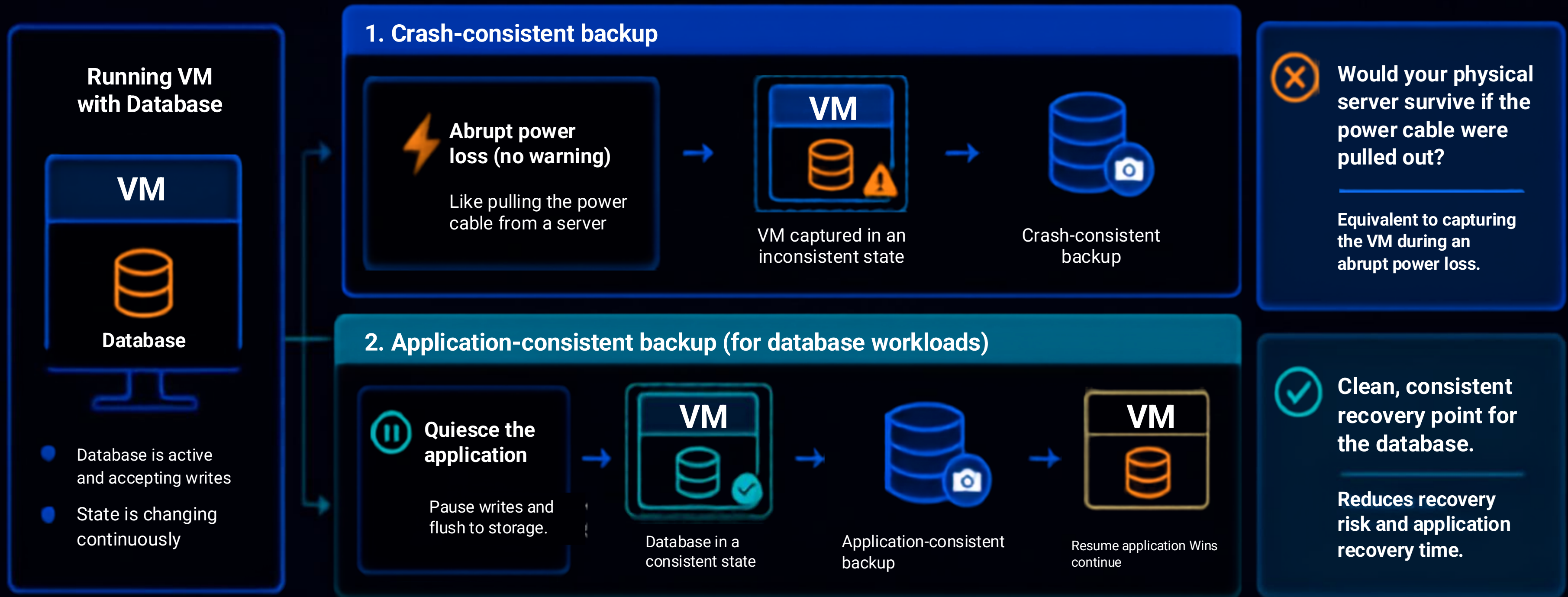
Destination: in-cluster

Namespace: kasten-io

Created At: 09/03/2026 13:49:48 (a day ago)

Last Sync: 09/03/2026 13:49:51 (a day ago)

Application Consistency Considerations



If the VM is running a database,
quiesce the application and include application consistency in the backup policy.

Up the stack, DevOps tooling protection

Accidents happen and attacks targeting accounts are also on the rise.

If you lost your GitHub account, what would you do?

1. GitHub account at risk



Your GitHub account could be compromised or accidentally lost.



2. What would you have to rebuild?

It's not just source code.
Recovery means rebuilding your entire GitHub platform configuration and collaboration history.



Rebuilding all of this is time-consuming, error-prone, and impacts the entire engineering organization.



3. Protect the whole pipeline!



Protect more than your repositories.

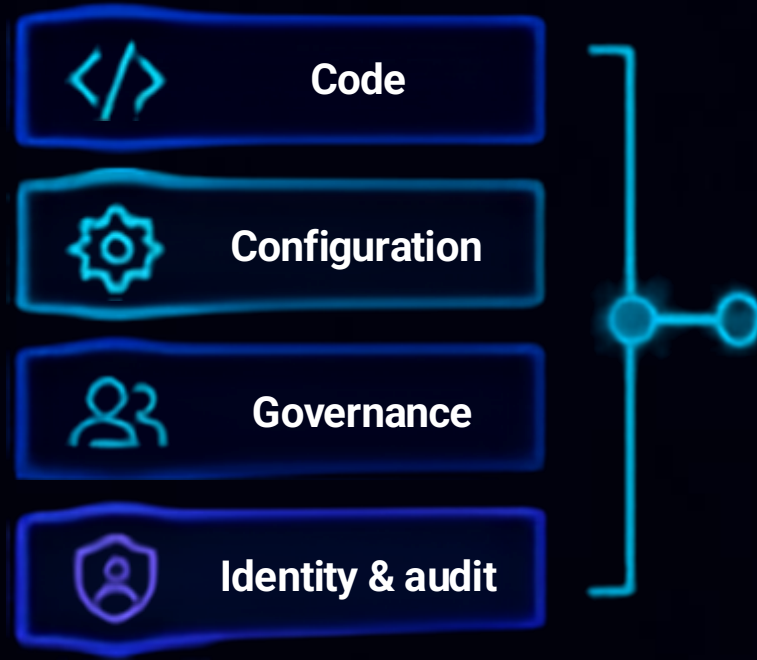
Protect your GitHub account and the full platform that powers your engineering collaboration.



Your DevOps platform is a critical asset. Protect the entire tool, not just the code.

We are working on something

1. Every layer, not just the code



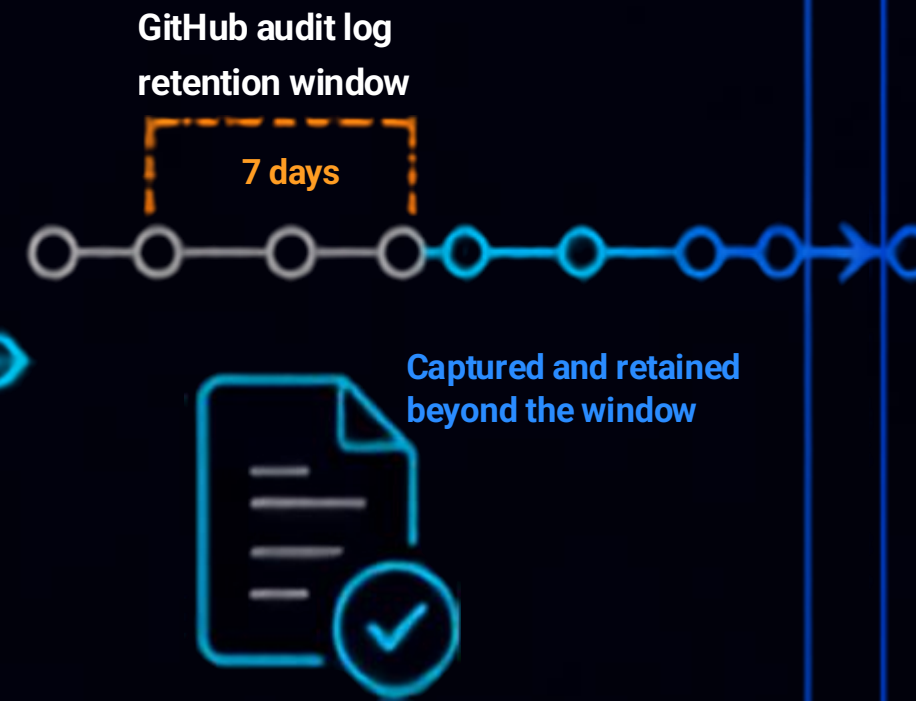
Configuration, Governance, Identity and audit — not only Git

2. Non-Destructive by rule, not by option



Nothing in production that is missing from backup is ever deleted, on any path.

3. Audit evidence that outlives the platform



GitHub deletes Git audit events after seven days. We will capture them before the window closes.

4. Not just GitHub!



The same approach across GitHub, Azure DevOps and GitHub



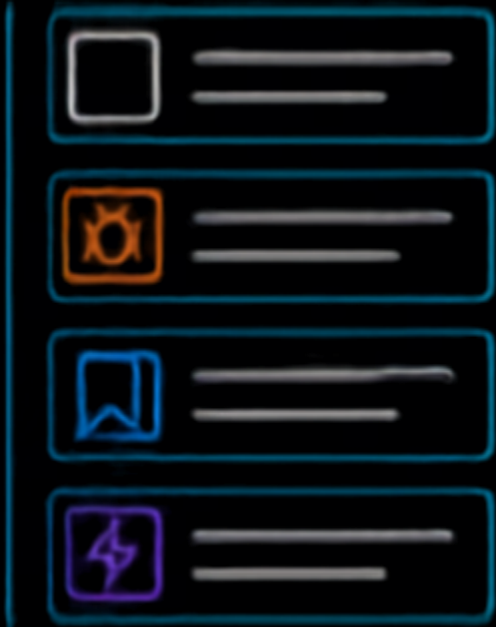
Expanding protection from code to the full DevOps platform — with evidence that lasts.

What about Atlassian Jira

Atlassian's Cloud issue-tracking and agile project management platform



1. Issues & Epics



Bugs, stories, tasks and epics — the atomic unit of engineering work

2. Workflows & Automation



Custom status transitions, approval flows, automation rules — often months of configuration work embedded here

3. Scrum & Kanban Boards



Sprint boards, backlogs and board configurations — the visual workflow layer teams interact with every day

4. Sprints & Velocity



Sprint planning, velocity tracking and burn-down charts — the historical record of delivery cadence and capacity

5. Audit & Compliance Logs



Change history, workflow transitions and field edits — the evidence trail required for SOX, HIPPA and ISO 27001



Protect more than issues — protect the full engineering work platform end-to-end.

Next Steps. Take Action.

Strengthen protection, automate recovery, and build resilience with GitOps.

01

Assess data protection gaps across VMs and containers

02

Implement ArgoCD-based declarative backup policies

03

Integrate DevOps tooling registries into your resilience strategy

04

Automate multi-cluster recovery and test failovers regularly

05

Explore Kubevirt and GitOps community resources

One more thing

Anyone struggle to keep/get an
inventory of all systems across all
platforms?

Multi-Cloud & Infrastructure
Discovery Made Simple



KOLLECT
CLOUD INVENTORY

- Single Binary
- CLI | Web UI | MCP
- Public Cloud, Cloud-Native, DevOps, Hypervisors

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